SingleStoreDB Technical Overview

Overview

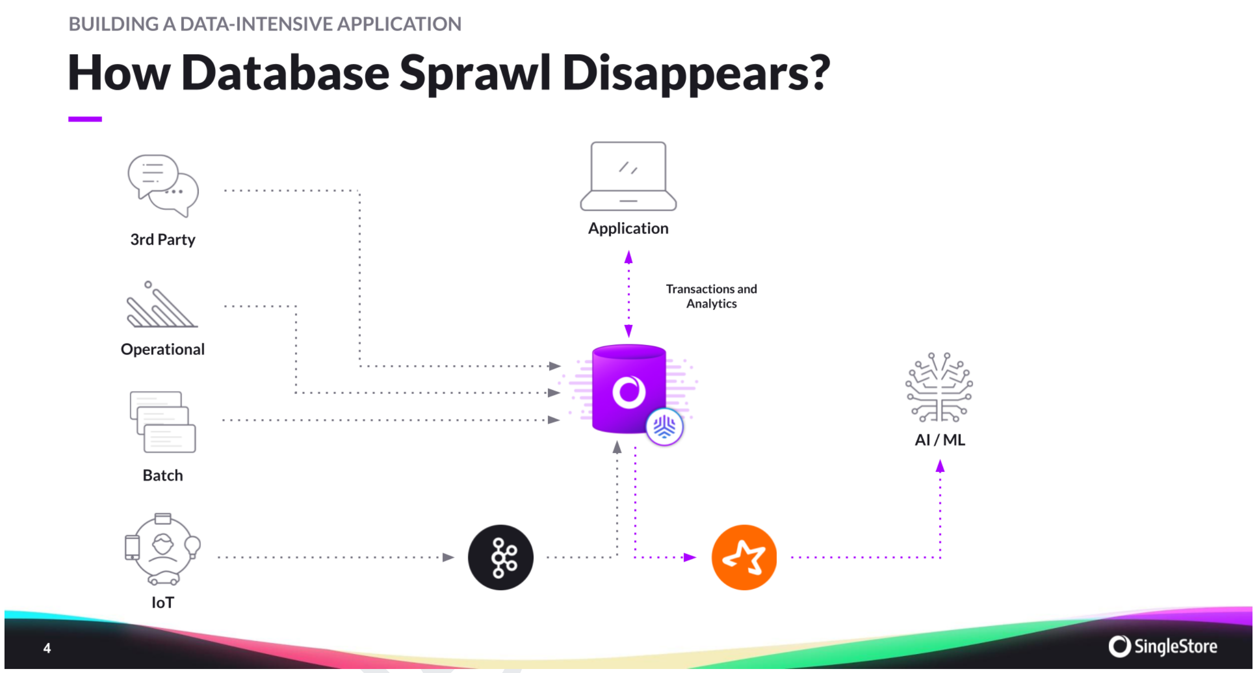
SingleStore DB is a cloud-native database designed for data-intensive applications. A distributed, relational, SQL database management system(RDBMS) that features ANSI SQL support. Best in class Database when it comes to speed in data ingestion, transaction processing, and query processing.SingleStore stores JSON data, graph data, and time series data besides relational data. It supports blended workloads, commonly referred to as HTAP workloads, as well as more traditional OLTP and OLAP use cases. The SingleStore database engine can be run in various Linux environments, including on-premises installations, public and private cloud providers, in containers via a Kubernetes operator, or as a hosted service in the cloud known as SingleStoreDB

- **Latency-Free Analytics**: SingleStore lets you achieve ultra fast query response with high concurrency across both live and historical data using familiar ANSI SQL
- **Ultra-fast Event-to-Insight Performance**: Deliver against the toughest service level agreements using parallel, distributed lock-free ingestion and real-time query processing
- **Scale Limitlessly:** Elastic scale-out architecture with distributed massively parallel data processing delivers consistent, predictable response under high ingest and user concurrency
- **Ease of Use and Flexibility:** SingleStore TM brings simplicity and ease to your data architecture by allowing OLTP and OLAP workloads to be processed on operational data using a single table type
- **Drop-in Compatibility:** Plug-in directly with existing technologies and skills with support for standard SQL, BI and distributed technologies like Amazon S3, Spark, Kafka and Hadoop
- **Powerful Programmability:** Supports simple, fast, and powerful in-database programming via MPSQL and an extensive set of built-in data types and functions

Solutions

❖ **Eliminate Database Sprawl**

Move beyond legacy data architectures that can't support multiple workloads and use cases, to a single database that unifies your data and supports multiple, massive workloads concurrently.



❖ **Accelerate Data-Intensive Applications at Scale**

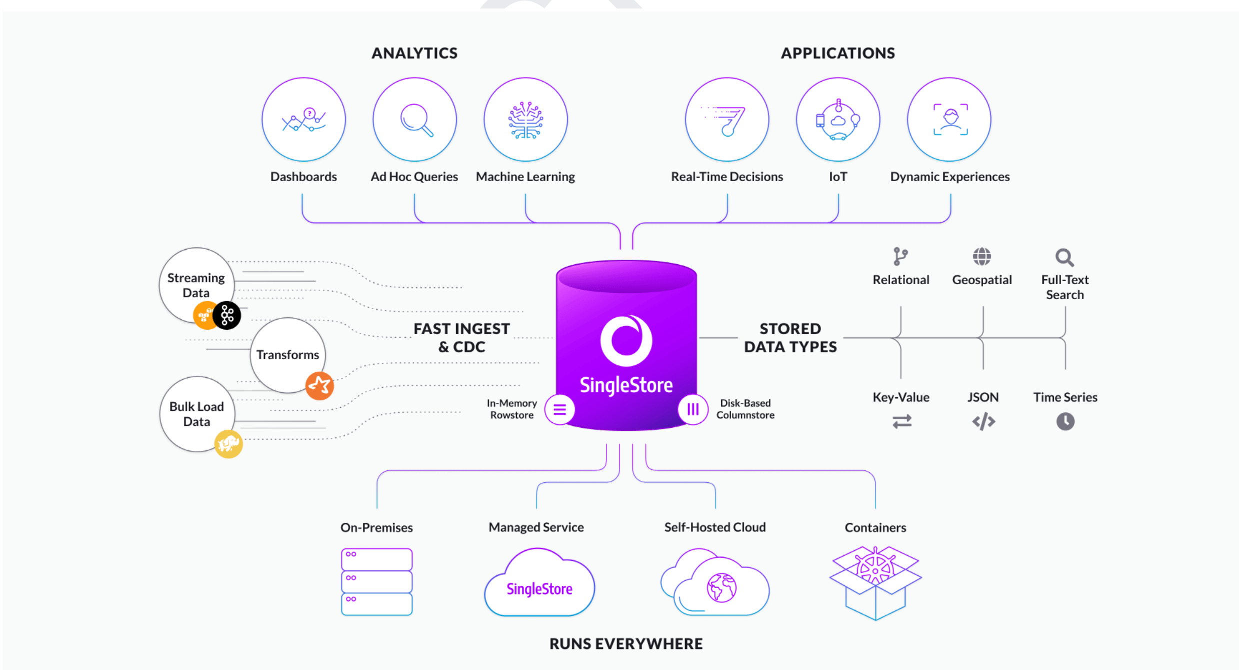
Power your applications with ludicrously fast modern database infrastructure that works on any data, anywhere, and at any scale.

❖ Deliver Real-Time, Ultra-Fast Analytics

Drive ultra-fast analytics on any data, anywhere, by leveraging our cloud-native, massively scalable database architecture that provides fast ingest and query performance with high concurrency.

❖ Accelerate and Simplify Data Ingestion

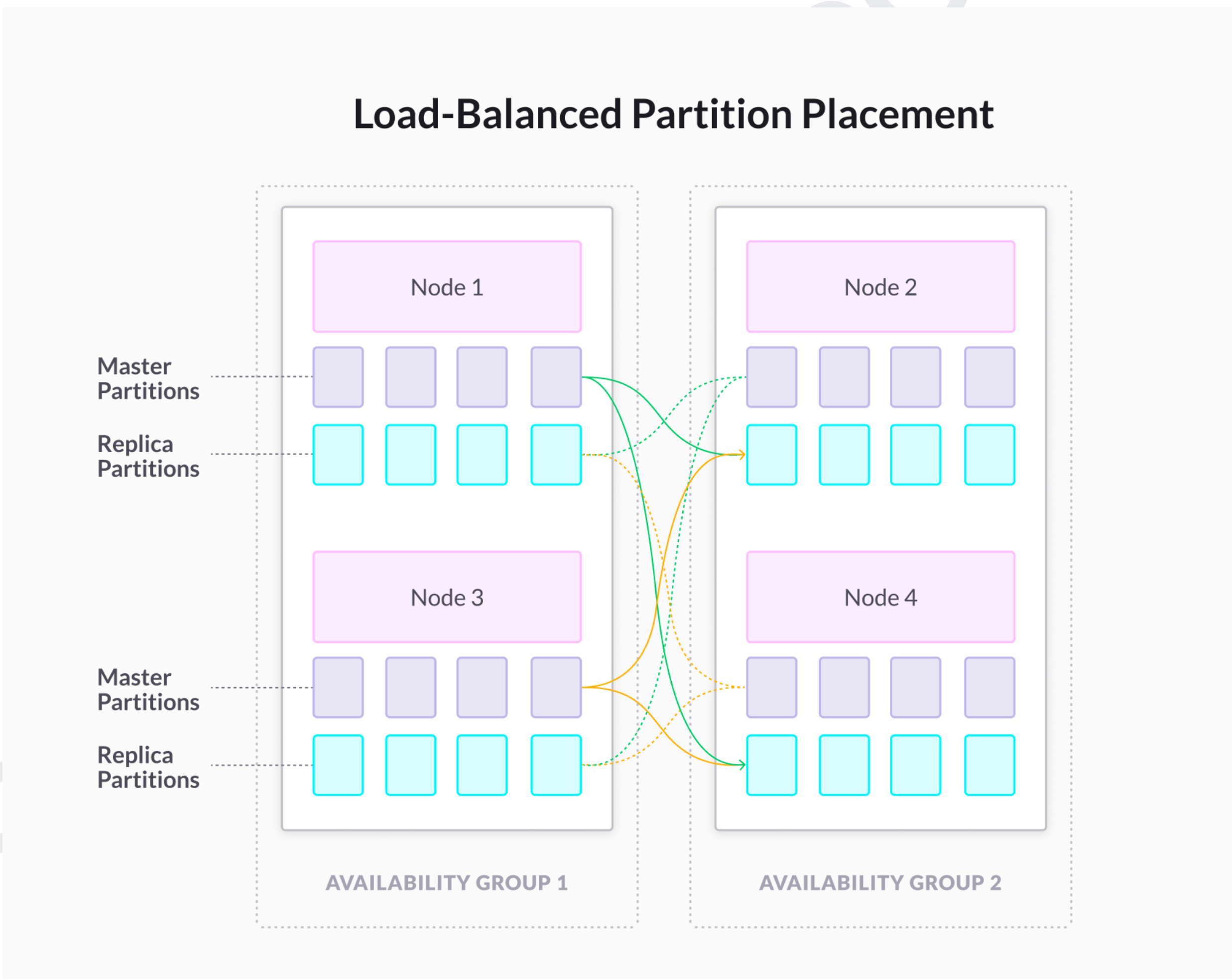
SingleStore is a distributed, highly scalable SQL database that can run anywhere. We deliver maximum performance for transactional and analytical workloads with familiar relational models



❖ Enhanced Reliability and Flexibility

Our highly-scalable, distributed system balances data and queries across a cluster of cloud instances or commodity hardware for maximum performance, concurrency, and availability

SingleStore is highly available by default. It ensures high availability by storing data redundantly in a set of nodes, called availability groups. SingleStore supports two availability groups. Each availability group contains a copy of every partition in the system—some as masters and some as replicas. As a result, SingleStore has two copies of your data in the system to protect the data against single node failure.



❖ Easily Deploy Clusters Anywhere

SingleStoreDB can be deployed anywhere in the cloud or on-premises without requiring specialty hardware to perform effectively; however, it will take advantage of single-instruction/multiple-data (SIMD) CPU instructions, specifically AVX2, and it is NUMA-aware to increase performance on newer machines. It can be deployed on bare metal, in VMs, and in the cloud.

❖ Comprehensive Security for Your Most Sensitive Data

SingleStore meets or exceeds data security requirements, including managing the most classified data, without compromising database performance. Easily manage how users and roles access data to support workloads in complex organizations and regulated environments.

Role-Based Access Control (RBAC)

Efficiently manage simple or robust security configurations by user role and group while maintaining maximum performance

Auditing

Database logging writes all activities to a secure external location to support information security tasks such as tracking user access.

Authentication

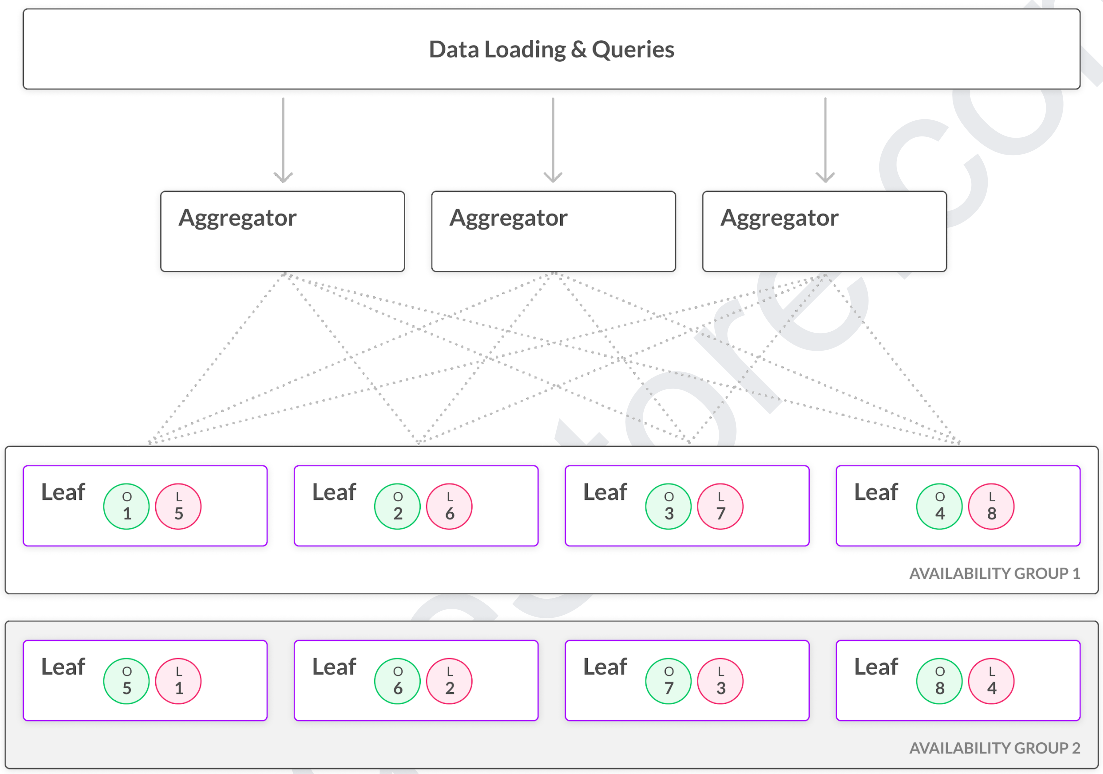
Manage existing account access with PAM (Pluggable Authentication Module) and SAML authentication support

❖ Security Certifications

SingleStore has secured industry-leading security certifications including ISO 27001 and SOC 2 Type 2, and is also fully compliant to the requirements of HIPAA, CCPA, and GDPR.

Architecture

SingleStoreDB stores and computes data on leaf nodes. You can linearly scale both storage and computational power by adding more leaf nodes. Clients query an aggregator node, which in turn queries one or more leaf nodes to collect the rows required to execute the query. Multiple aggregator nodes perform the same functions with respect to executing Data Manipulation Language (DML) queries and allow clients to load-balance queries across the aggregators. Leaf nodes should not be queried directly except for maintenance purposes in exceptional situations.



Referential Links

- **eLearning: training.singlestore.com.**
- **Documentation: Read through topics at docs.singlestore.com.**
- **Blogs: Read through SingleStore blog posts at singlestore.com/blogs.**